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Assessing public transport accessibility using GPS data

Martin Haamer^{1*} , Anto Aasa¹ and Age Poom¹

Abstract

The transition to sustainable urban mobility emphasises the need for an equitable public transport system that gives users high levels of access to their desired destinations. Critical to this effort is a comprehensive understanding of urban travel patterns and needs. Mobile positioning technologies such as GPS enable the collection of large-scale, detailed data on human mobility, where realised travel behaviour functions as an indicator of urban travel needs. However, GPS data has been underutilised in accessibility studies, either due to unavailability, the conceptual gap between potential and realised travel, or methodological challenges in transforming large-scale datasets into meaningful accessibility indicators. Part of these challenges includes identifying individual trips from continuous GPS data, a process known as trip chain detection. Furthermore, despite accessibility being an inherently dynamic concept, the temporal dimension is often ignored in public transport accessibility studies. The contribution of this paper is twofold. First, we introduce a novel methodological framework for trip chain detection from continuous GPS data without supplementary information. Second, we demonstrate how GPS data can reveal realised travel to help prioritise destinations for accessibility studies, making accessibility metrics more relevant to real-world travel needs. To do so, we incorporate a dynamic accessibility approach to account for the variation in travel time at different times for a comprehensive understanding of public transport accessibility. We model public transport and walking accessibility using General Transit Feed Specification (GTFS) data in a GIS environment. The methodological approach is exemplified in the case of public transport accessibility in Tallinn, Estonia. Our proposed methodology enables transport planners to consider realised travel behaviour and origin-destination locations in public transport-related decision-making to enhance spatiotemporal accessibility of our desired destinations.

Keywords Public transport, Dynamic accessibility modelling, GPS, Trip chain detection, Urban mobility

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1 Introduction

The European transportation sector is responsible for approximately one-quarter of the EU's total greenhouse gas (GHG) emissions, with road transport, particularly passenger cars, accounting for the largest share (71.7% and 43.5% of sector emissions, respectively, in 2019) [10]. Reducing the carbon footprint of transportation is considered one of the main challenges in the transition to a sustainable future [12]. Shifting travel behaviour towards sustainable modes, such as walking, cycling, and public transport, can substantially reduce car dependence and resulting GHG emissions in cities. Realising these benefits requires transport networks that enable high levels of access to desired destinations by sustainable modes. The concept of accessibility was first defined by [19] as "the potential of opportunities for interaction" and has since evolved to encompass various definitions representing the differences in ease of access to locations [26]. Common among all definitions is the focus on the ability to reach destinations, activities, or opportunities by different means of transport [2]. Accessibility is inherently both a spatial and temporal phenomenon. Yet, including time in spatial accessibility modelling has been limited [23]. This can be attributed to the previous lack of detailed temporal data on public transport services and increased complexity in modelling [41]. Ignoring the accessibility of destinations and fluctuations in travel times throughout the day results in an incomplete understanding of accessibility, valid only at a fixed moment in time. Mapping the variations in accessibility is crucial in guiding transport planning to improve public transport services and thereby enhance spatial accessibility.

Detailed data on both public transport services and travel behaviour are essential in assessing the efficiency of a public transport system. The widespread use of location-based technologies such as GPS, combined with advancements in computational power, has allowed for detailed analysis of individual mobility patterns over extensive periods of time [4]. One of the fundamental challenges in using GPS data in mobility studies is identifying periods of movement and stationary activities, also known as trip chain detection.

In this paper, we use long-term GPS data to study desired travel destinations and analyse public transport accessibility to the identified destinations. For public transport accessibility modelling, we use General Transit Feed Specification (GTFS) data in a GIS environment. The contribution of our research is twofold. First, we demonstrate a novel, straightforward and intuitive methodological framework for identifying trips and destinations from continuous GPS data without requiring supplementary information from the respondent. Second, we apply the results from this analysis to evaluate public transport accessibility to destinations revealed

from GPS data. This enables accounting for real-world travel needs when calculating accessibility metrics and hence overcome the eventual mismatch between travel demand and public transport provision in transport planning. We incorporate a dynamic accessibility approach to account for the variation in travel time at different times for a comprehensive understanding of public transport accessibility. This knowledge provides further information for targeting improvements in public transport accessibility with maximum impact.

The paper is structured as follows. In chapter two, we provide an overview of the strengths and shortcomings of several approaches used to assess public transport accessibility. We also discuss the advantages and challenges of using GPS data for mobility studies. In chapter three, we explain our methodological approach and its application in the case study area of Tallinn, Estonia. We outline the results in chapter four and conclude our paper in chapter five with a discussion of the main findings, limitations, and further areas of research.

2 Literature overview

2.1 Public transport accessibility

One of the main aims of transport policy is to improve access to desirable destinations and activities, thereby increasing social equity [22]. Accessibility measures capture the ease with which destinations can be reached from different locations, considering the quality and availability of transportation infrastructure [30]. Poor public transport service can restrict access to destinations and consequently lead to transport-related social exclusion [3]. Measuring accessibility levels by public transport and identifying poorly connected areas is, therefore, a core element in addressing transport disadvantage [33].

Compared to other modes of travel, accessibility studies focusing on public transport are limited. The reasons include complicated modelling due to the specificities of public transport, and the scarcity of necessary data on service provision in the past [29]. According to [29], most public transport accessibility studies use one of three approaches. They either measure (1) access to transit stops; (2) the duration of the public transport journey; or (3) access to destinations via public transport. The first approach calculates physical access to the public transport network by measuring the proximity to transit stops in time or distance. While the approach is commonly used due to its simplicity and easy computation, it disregards the level of available service and route network from the transit stop [29, 41]. The second approach accounts for in-vehicle travel time in addition to the time accessing the transit stop. This enables mapping accessible areas from a certain location by public transport and walking within a fixed temporal threshold [28]. However,

this method focuses solely on travel time without considering the value of destinations. Building on the previous, the third approach evaluates accessibility with regard to the availability of destinations and the types of activities available within the areas accessible.

These approaches can be advanced by a fourth, the most sophisticated approach, that additionally accounts for the temporal variability in public transport accessibility [15]. Studies measuring accessibility based on travel time have typically focused on specific times of day, particularly peak hours when transport services run most frequently [29]. Relying solely on peak hour schedules while ignoring the temporal variability of services throughout the day tends to overestimate the accessibility typically perceived by users [13, 41]. The widespread availability of detailed schedule-based travel time information in the form of GTFS datasets has enabled studying time-of-day variations in public transport accessibility at a large scale [17, 41, 46]. These studies use detailed door-to-door approaches accounting for multimodality in accessing destinations [14, 15, 35, 42]. However, dynamic public transport accessibility has remained understudied, as including the temporal dimension increases complexity and imposes a heavy burden on computational power since the analysis must be repeated for every moment in time [41].

Our study accounts for the variability of access based on the time of day when analysing public transport accessibility. This approach helps understand the spatial differences in access to destinations throughout the day, mitigating the risk of social exclusion. Furthermore, we also account for how frequently people visit destinations. It is equally important to differentiate the destinations people actually go to from the range of places reachable [33].

2.2 Large-scale mobility data in travel research

Travel behaviour studies that reveal mobility needs and patterns have traditionally been based on user-reported methods such as questionnaires, recall interviews, and travel diaries [43]. These approaches either trade off on sample size or level of detail [4], and a persistent challenge in travel behaviour research until recent decades has been the lack of large-scale, detailed data on people's movements [45]. Estimating origin-destination flows based on large-scale observations of travel patterns has only become possible with the emergence of passively collected big data, such as mobile phone data [6]. One of the main advantages of mobile phone data is the automated collection of large amounts of data on spatiotemporal trajectories and travel origins and destinations [16]. Cellular network-based mobile phone data in the form of call detail records (CDR) have been the most commonly used large-scale data in travel behaviour research [45].

For example, [1] used CDR data to study the differences in the temporal movement patterns of suburban commuters in Tallinn, Estonia.

GPS data, as another large-scale spatiotemporal data type collected either by mobile phones or specific devices, offers the most potential for understanding detailed travel behaviour, but is yet to be adopted by most stakeholders in transportation policy and planning [20]. Compared to traditional travel data collection methods, GPS-based studies provide higher spatial and temporal accuracy and reduce the response burden for participants [36–38]. These benefits reduce the possibility of errors made by the respondent in trip logging and enable the study of individual mobility patterns over long periods.

One of the fundamental challenges in using GPS data in travel research is the lack of additional data, e.g., on trip purpose and travel mode, unless specifically requested from the participant [20]. This necessitates extensive data processing to enable mode estimation and use in more sophisticated analysis [36]. Trip chain detection from continuous GPS data is a persistent challenge, exacerbated by the innate variability of GPS accuracy, as it depends on the surrounding environment and satellite availability. Additionally, accurately classifying GPS data segments as either stops or movements remains challenging, and the limited transferability of methodological approaches across studies has hindered the widespread use of GPS data in transportation research [9].

Large-scale mobility data collected over longer periods of time have seldom been used to analyse spatial accessibility by public transport and the mismatch between transport demand and supply. Most studies have relied on various datasets to estimate travel demand, for example, land use and population density data [49], the spatial distribution of socio-demographic groups [5, 15], census data [24], or locations of workplaces [18]. Our study is one of the few to use long-term GPS data to analyse travel demand, reveal destination preferences, and evaluate public transport accessibility to desired destinations while accounting for temporal variability throughout the day.

3 Data and methodology

3.1 Workflow

Our research consisted of two distinct yet interconnected analyses. The first analysis focused on identifying trip destinations by constructing an origin-destination (OD) matrix from continuous GPS data. The data were collected from residents in two case study areas in Tallinn, Estonia (see Sect. 3.2). This involved creating a novel framework for algorithm-based trip chain detection from GPS data without supplementary data on the respondent. The second analysis investigated how public transport accessibility to destinations observed from GPS data

varied throughout the day. The methodology used in our study is summarised as a flowchart in Fig. 1.

3.2 Study area

The methodology was developed and tested based on data collected from Tallinn, the capital of Estonia. Tallinn is the most densely populated region in the country, with 457,572 inhabitants [39]. The region has rapidly grown during the last decades and has seen ever-increasing car use since the 2000s. Despite public transport in Tallinn being free for its residents since 2014, the mode share of public transport has steadily decreased, with a slight increase in recent years following the COVID-19 pandemic (Fig. 2) [44]. 49% of the working population commutes primarily by car, and 31% use public transport [44]. Tallinn's mobility strategy aims to increase the share of movement by public transport, on foot and by bicycle from 41% in 2020 to 70% by 2035 [7].

We set the spatial extent of our study area based on the administrative borders of Tallinn. We divided the city into 270 transport analysis zones as described by [25] (Fig. 3). The area of each zone varies depending on population density. The average zone consists of 1,704 residents and covers an area of 0.58 km².

Two case study areas, Kalamaja and Priisle neighbourhoods in Tallinn, were used for GPS travel data collection among their residents (see Sect. 3.2 and 3.3). These neighbourhoods were selected to represent contrasting urban environments, socioeconomic characteristics of the population, and residents' mobility patterns within Tallinn. Kalamaja neighbourhood is a predominantly Estonian-speaking gentrified neighbourhood in northern Tallinn, located close to the city centre (Fig. 3). Kalamaja exemplifies a well-connected inner-city area where nearly half of the residents (45%) exhibit walk-dominant mobility behaviour [37]. In contrast, Priisle neighbourhood is a dense, predominantly Russian-speaking Soviet-era

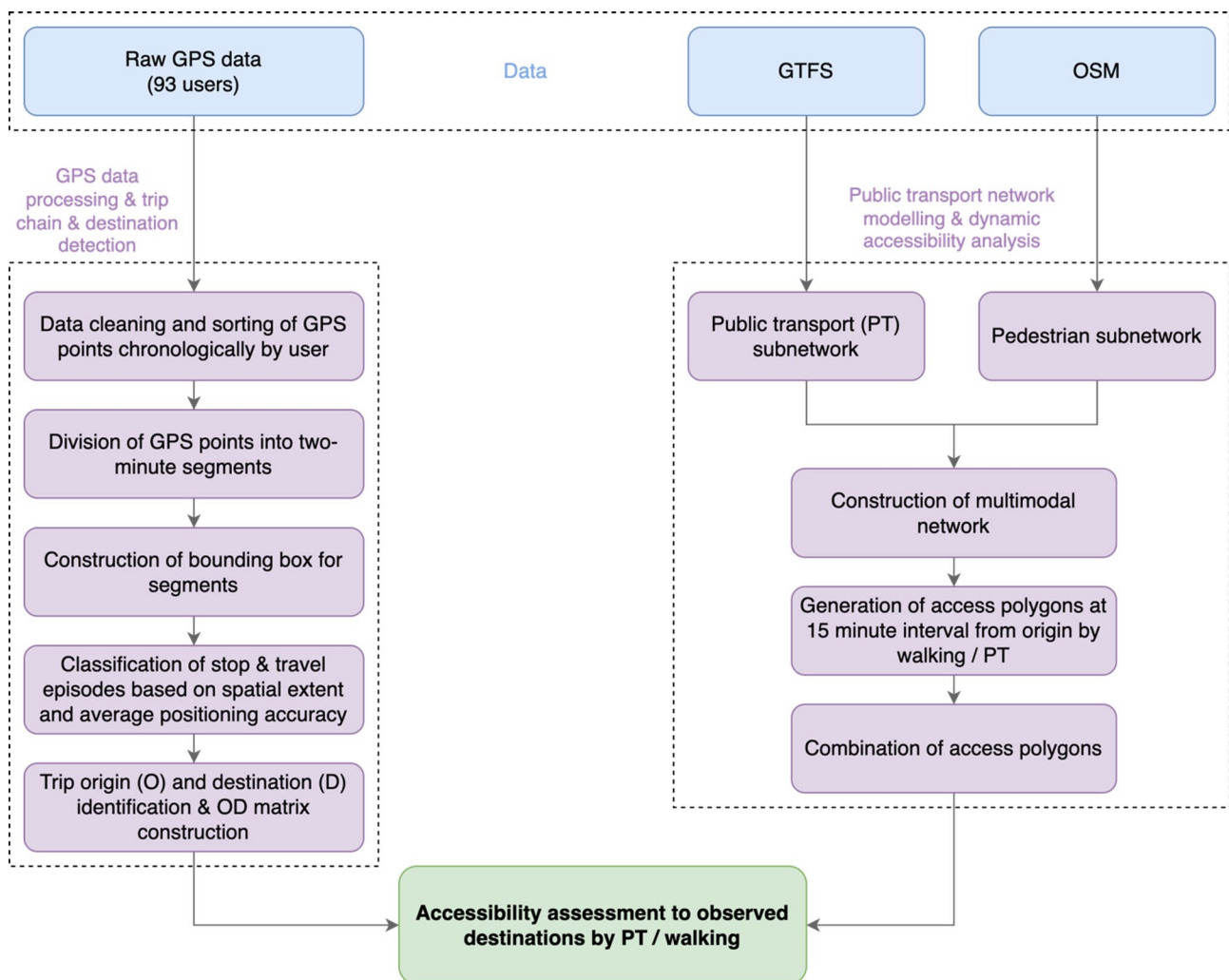


Fig. 1 Flowchart of the study methodology. GTFS – General Transit Feed Specification. OSM – Open Street Map

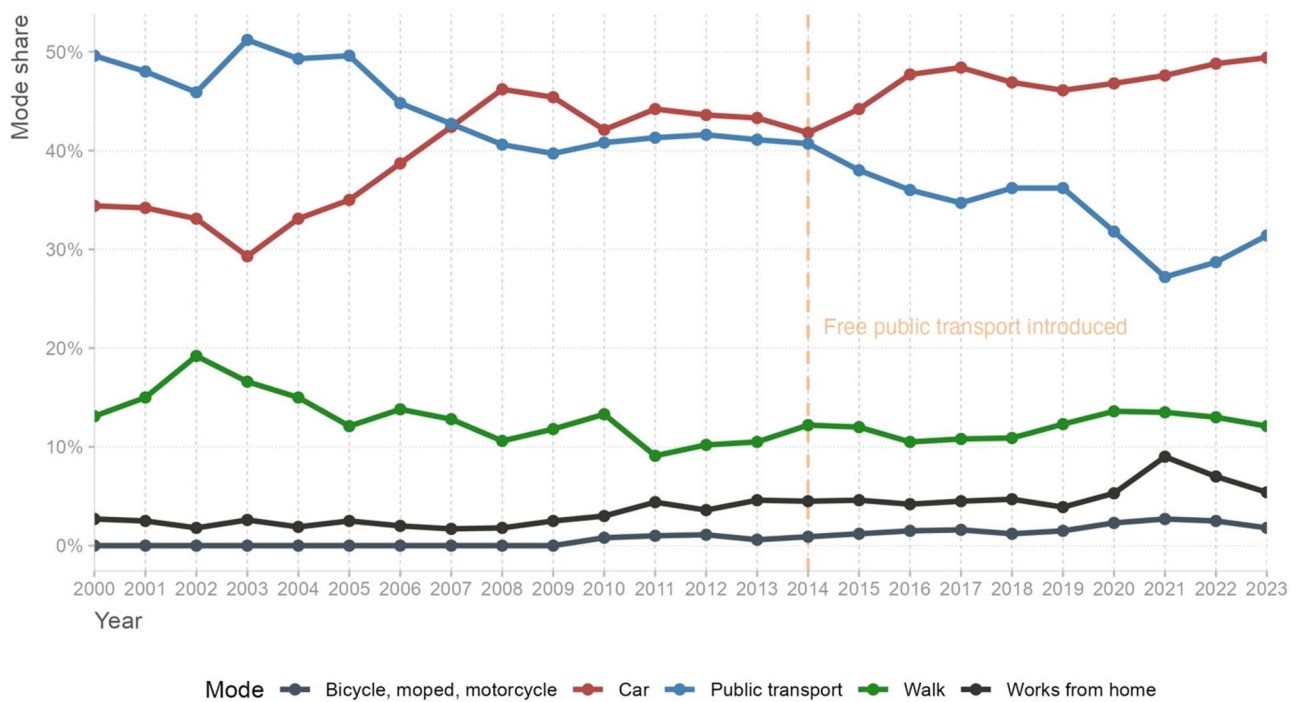


Fig. 2 Modal split in Tallinn for commuting from 2000 to 2023. Source: Transport Administration, [44]

neighbourhood with large housing estates on the outskirts of eastern Tallinn. Priisle represents a peripheral suburb where car use predominates (38% of residents), though a significant proportion (21%) relies primarily on public transport [37].

3.3 Data and sources

The data used in our study have been summarised in Table 1.

3.3.1 3.3.1 GPS data

We used mobile phone-based GPS data on daily mobility from a sample of 93 individuals from 01.09.2021 to 01.12.2021 to detect trip destinations (Table 2). Informed consent was obtained from all participants, and the study was approved by the Research Ethics Committee of the University of Tartu (approval no. 298T/12). The sample was created for a separate research project funded by the Estonian Research Council (PRG306) and designed to reflect the balance of various social statuses in alignment with their distribution in the population of Tallinn according to the 2021 national census [37]. Sample creation followed quotas for age, gender, and mother tongue and consisted of adults aged 25 to 65 (mean age of 44). The sample was unequally divided between the two case study areas — Priisle neighbourhood (56 individuals) and Kalamaja neighbourhood (37 individuals). GPS data was collected by a purpose-built app, MobilityLog, developed by the Mobility Lab at the University of Tartu [34]. The dataset consisted of over 22 million GPS point locations

collected across 5,806 user-days, representing the total number of days users contributed their location data. The number of user-days per user ranged from 6 to 92, with a median of 72 and a standard deviation of 31.2.

3.3.2 MobilityLog app

MobilityLog is an Android system-based smartphone application that collects user location and phone usage data. The app gathers GPS coordinates, timestamps, estimated accuracy, bearing, and speed between the coordinates. The app only collects a user's location data in case of movement. Data collection starts if movement is detected three times within 10-second intervals, and the frequency of data collection adjusts based on speed: at speeds under three m/s, data is recorded every 16 s; at higher speeds, it is recorded every second. Given the high energy demand of satellite positioning [16], this adaptive approach minimises battery consumption while maintaining spatial accuracy. Data collection is temporarily paused if no movement is detected for 30 s or more and resumes if movement is detected or the Wi-Fi network changes. Every five minutes, the app checks for new movement signals. Once collected, the data is compressed, encrypted, and automatically uploaded to a secure server. An automated data processing system ensures the data is cleaned of errors [34].

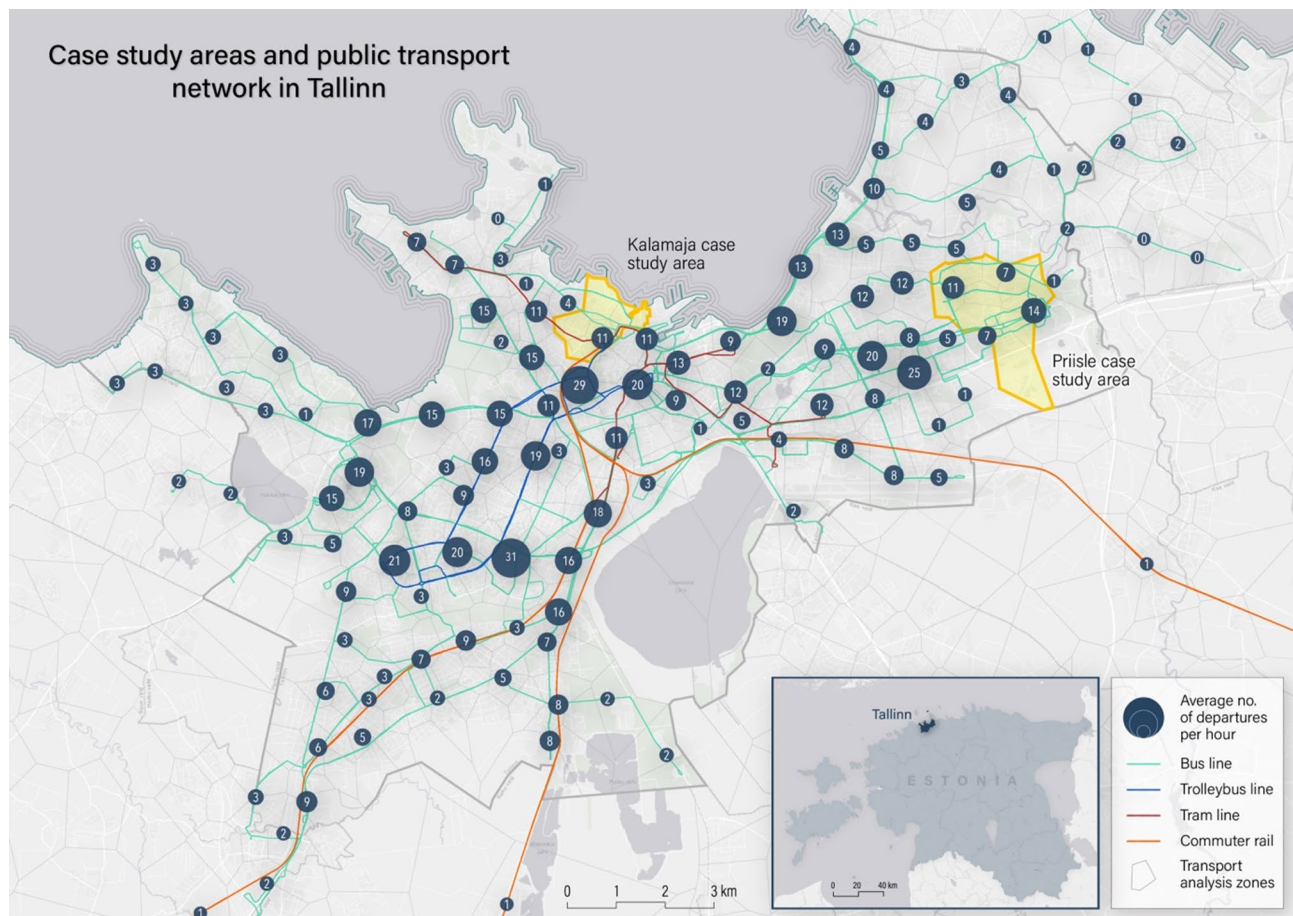


Fig. 3 Case study areas Kalamaja and Priisle, transport analysis zones, and public transport network in Tallinn

Table 1 Data sources

Data	Source	Description
GPS data	MobilityLog smartphone application	GPS data from residents of Kalamaja and Priisle. Collected from 93 people between 01.09.2021 and 01.12.2021.
Public transport data	Estonian Transport Administration	GTFS data on services operated by Tallinna Linnatranspord and Elron. Data obtained 23.02.2022. (https://peatusee/gtfs/)
Street network	OpenStreetMap	Spatial features, such as lines, are used for constructing the pedestrian network. Data obtained 23.02.2022. (https://download.geofabrik.de/europe/estonia.html .)

4 Network datasets

We used two network datasets to model public transport accessibility to destinations observed from GPS data. The first dataset was a logical, graph-based network representing transit stops, routes connecting the stops, and corresponding schedules for different modes of transport. This dataset was used to model in-vehicle travel times and was constructed using GTFS data, which has become the de facto standard for publishing public

Table 2 Sample distribution among socioeconomic characteristics

Category	Group	Kalamaja	Priisle
Gender	Female	24 (64.9%)	32 (57.1%)
	Male	13 (35.1%)	24 (42.9%)
Age group	...30	8 (21.6%)	5 (8.9%)
	31...40	7 (18.9%)	16 (28.6%)
	41...50	8 (21.6%)	15 (26.8%)
	51...	14 (37.8%)	20 (35.7%)
Education	Higher	31 (83.8%)	26 (46.4%)
	Lower	6 (16.2%)	30 (53.6%)
Income	High	13 (35.1%)	10 (17.9%)
	Mid	10 (27.0%)	27 (48.2%)
	Low	7 (18.9%)	10 (17.9%)
	No answer	7 (18.9%)	9 (16.1%)
Language	Estonian	26 (70.3%)	19 (33.9%)
	Russian	7 (18.9%)	34 (60.7%)
	Other	4 (10.8%)	3 (5.4%)

transport information [46]. GTFS datasets consist of several comma-separated text files with relational dependencies between the files describing the operational aspects of the public transport network.

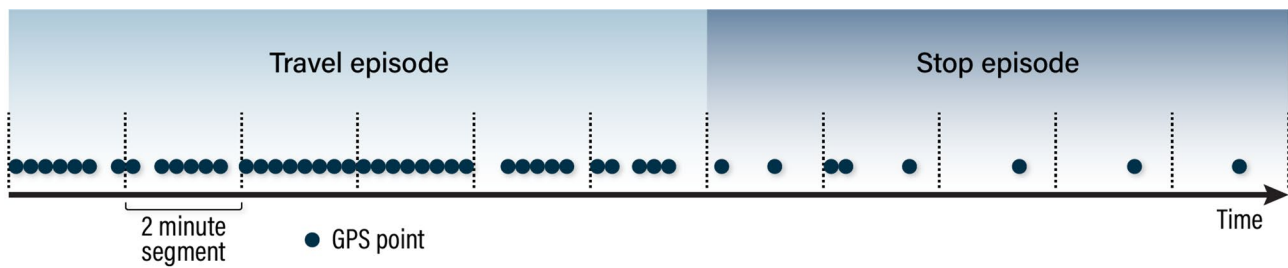


Fig. 4 Division of the user's sequential GPS points into two-minute segments

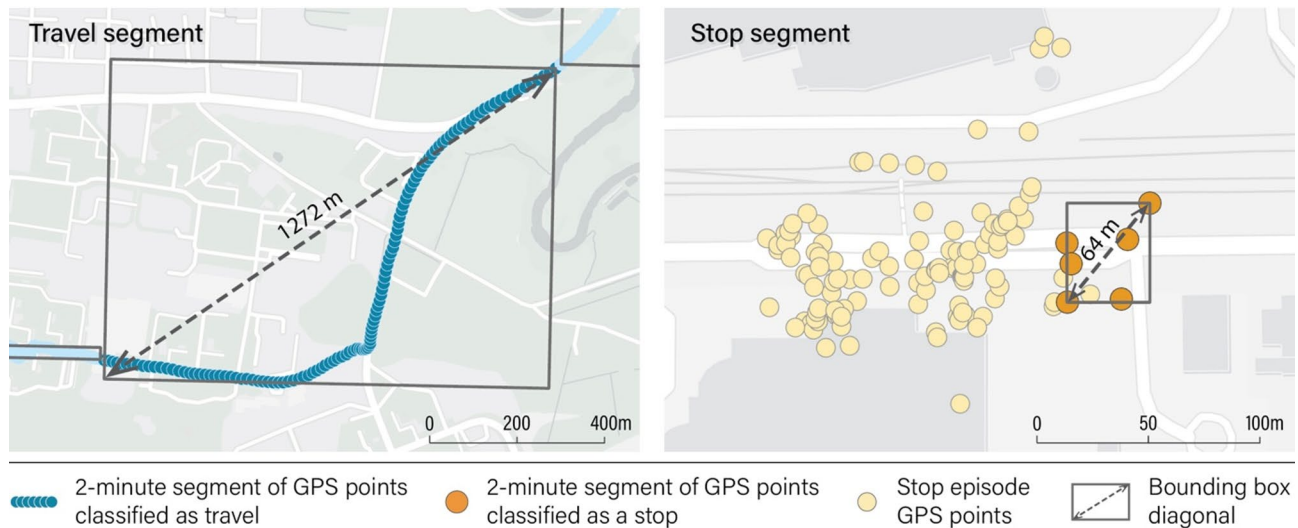


Fig. 5 Example of classifying two-minute segments as travel or stop segments based on the diagonal length of their bounding box

The second dataset was a spatially accurate street network sourced from OpenStreetMap (OSM), used to model pedestrian access to public transport stops and destinations. We applied restrictions on links unsuitable for pedestrian travel (e.g. highways and major roads). We used GIS software to validate the network's topology, ensuring it remained navigable despite the imposed restrictions and modified the restrictions on certain links accordingly. We added additional links based on local knowledge near train stations to ensure all access points were accurately represented in the network. Following the findings of [40], we adopted a pedestrian walking speed of 4.5 km/h for our model.

5 Methodology

Our study combines GPS-derived origin-destination matrices with simulated public transport accessibility to analyse the mismatch between observed travel demand and the accessibility provided by the public transport system. GPS data were used to identify individual trip episodes and their origins and destinations, revealing prevalent trip destinations. Separately, we modelled public transport accessibility across the study area, using a realistic door-to-door framework accounting for

all stages of the journey. By comparing the destinations to which people travel with their accessibility by public transport using the same transport analysis zones, we assessed the spatial and temporal mismatch between travel behaviour and transport service provision.

6 Trip chain detection

We developed a novel approach for detecting stationary activities and non-stationary travel episodes from continuous GPS data. First, we divided each user's sequential GPS points into two-minute segments, based solely on their timestamps (Fig. 4). For each segment, we calculated spatial and descriptive statistics, including distance covered, spatial extent (based on the segment's bounding box), and average positioning accuracy, measured by the dilution of precision (DoP) [27].

Second, we classified travel segments if the spatial extent exceeded 100 m, and its average theoretical positioning accuracy was better than 20 m. Otherwise, the segment was classified as a stop segment (Fig. 5). We set the minimum duration of a travel episode to four minutes (i.e., at least two consecutive travel segments). Hence, any travel episode lasting less than this threshold was reclassified as a stop episode.

Third, we combined consecutive travel segments into a single travel episode (trip). The centroid of the first segment defined the zone of origin, and the centroid of the last segment defined the destination zone. If the elapsed time between two consecutive trips was under 10 min, the trips were merged to ensure different stages of a single trip were not misclassified as separate trips. Finally, we only retained trips with both origin and destination within our study area. Trips where the distance between the start and end points was under 500 m were also removed. Altogether, 8917 trips were identified from the dataset using this approach (Fig. 6).

We found that dividing GPS traces into two-minute segments was suitable for trip chain detection, given the average zone size in the case study area and the data collection interval of the MobilityLog app. The segment duration is crucial in accurately identifying the zones of trip origin and destination, as it depends on the spatial unit of analysis and the temporal resolution of the GPS data. Suppose the chosen period is too long, short intermediate destinations risk being undetected. The chosen two-minute durations proved effective, as a pedestrian with an average walking speed of 4.5 km/h will traverse

150 m within this time, which is sufficient in differentiating travel episodes from stationary activities.

After testing multiple threshold values and visually assessing the classification accuracy of trip chain detection in ArcGIS, we identified a 100-meter threshold as optimal for distinguishing between stop and travel episodes. This distance effectively captured travel and stop episodes, ensuring that brief pauses or short trips were accurately classified. The trips were aggregated by the transport analysis zones of their origins and destinations to allow for a consistent spatial unit in analysing public transport accessibility to the identified destinations. While GPS data would enable destination analysis at a finer spatial scale, the limited sample size in our study limited our analysis to the zone level. Additionally, trips occurring entirely within the case study areas were excluded from the dataset. Altogether, 5017 travel episodes (trips) were identified, and ranged from 10 to 424 per user, with a median of 88.5 and a standard deviation of 91.9.

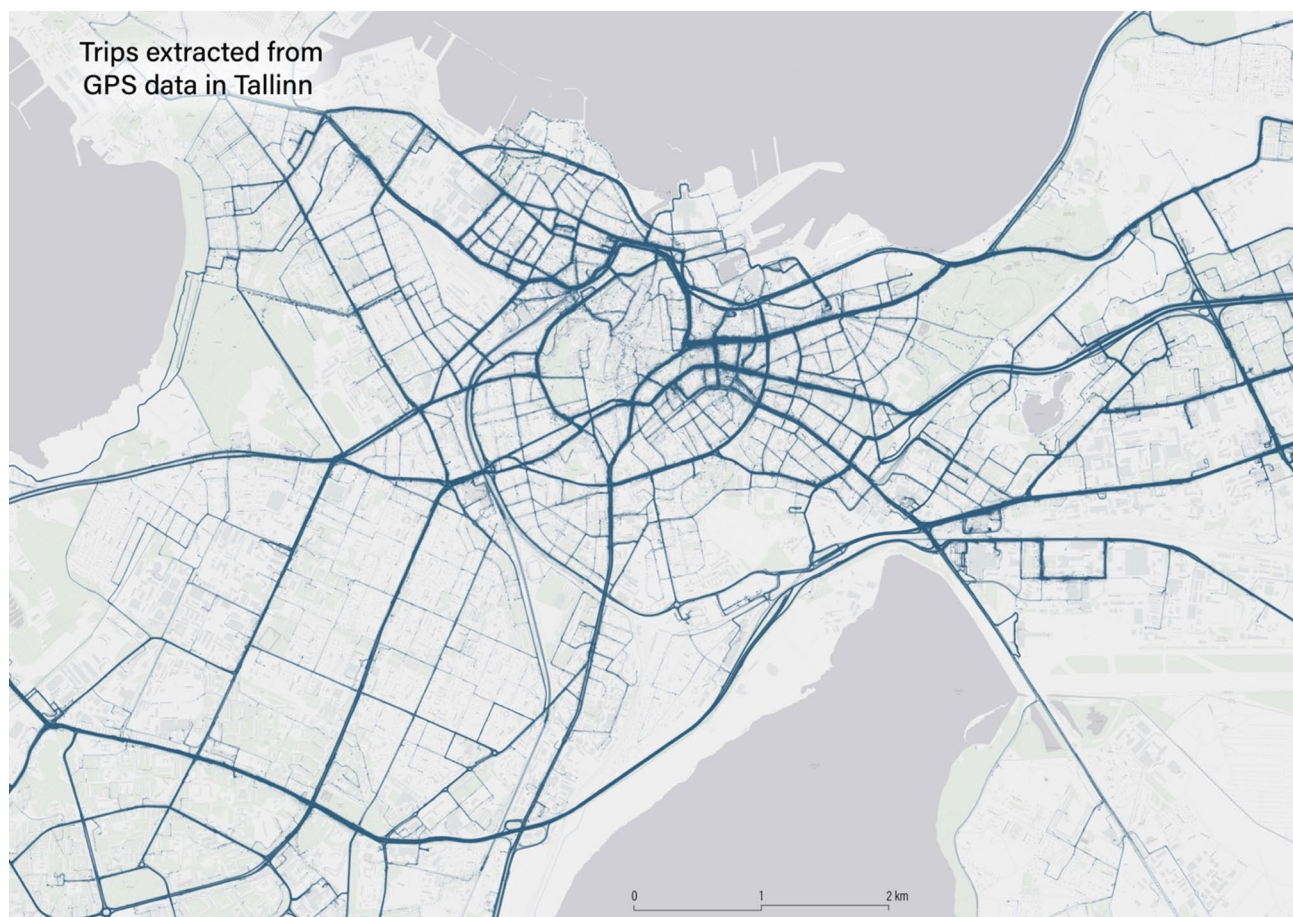


Fig. 6 Trips identified from GPS data in the city centre of Tallinn

7 Assessing public transport accessibility

We constructed a multimodal transport network in ArcGIS using the Network Analyst extension developed by ESRI to simulate travel between locations via walking and public transport (PT) [21, 41]. The simulations were based on a network dataset — a structured graph of nodes and edges. The modelled network consisted of two distinct subnetworks: (1) a network of edges and nodes representing the PT services between stops; and (2) a topologically accurate pedestrian network. PT stops served as connectors between these subnetworks, enabling the use of both subnetworks and respective travel modes in modelling trips between any two locations. Our model enabled door-to-door accessibility assessment, incorporating all trip segments like journey planners. This included walking time from the zone centroid to the nearest PT stop, waiting time at the stop, in-vehicle travel time through the PT network, and walking time from the final stop to the destination zone centroid. Transfer times between PT services — including walking between stops — were also included. Although we used zone centroids as proxies for trip origins and destinations, the same approach can be adapted to model trips at a finer spatial resolution from individual buildings within a zone. Each zone was represented by its geometric centroid, which served as the reference point for all trips associated with that zone. We verified that the locations of these centroids were accessible within the network dataset and did not fall on bodies of water, adjusting their positions within each zone if needed. The case study areas, Kalamaja and Priisle neighbourhoods as home locations of the study participants, included five and nine zones, respectively. For both case study areas, we used the population-weighted mean centre of the zones in the case study areas as a proxy for trip origins.

By calculating the accessibility by public transport and walking from the case study areas of Kalamaja and Priisle, we identified the zones reachable within 30 min from each case study area's mean centre. This 30-minute threshold aligns with Tallinn's mobility strategy, which aims to limit average public transport journey times to within half an hour [7]. To assess the variations in

accessibility throughout the day, we repeated the analysis at regular intervals. For our temporal scope, we defined a “day” as 06:00 to 23:45, which aligns with the frequent public transport services period. We repeated accessibility assessments every 15 min. Stępnik et al. found this temporal resolution to provide the optimum balance between precision and computational time [41]. In total, we calculated 72 separate areas of access and overlaid the results to analyse changes in coverage across the day. We categorised the realised destinations into six levels of access (levels 0–5) by calculating the percentage of times each was reachable within a 30-minute journey throughout the day. Level five represented consistent access throughout the day (over 80% of the time), and level zero represented no access at any time of day. While multiple travel options typically exist for each origin-destination pair, the routing engine prioritises the fastest travel time regardless of the mode of travel, selecting walking when it is quicker than public transport. Accessibility calculations were executed using ESRI's Transit Network Analysis Toolkit [11].

8 Results

Table 3 summarises our results of public transport accessibility to observed destinations from GPS data.

Figure 7 depicts dynamic public transport accessibility during a typical workday from the Kalamaja neighbourhood within a 30-minute journey. Of the observed destinations, 86.5% are accessible within half an hour at least once during the day, whereas 74.6% are accessible throughout the day (over 80% of all times). High levels of dynamic public transport access are provided in several geographic directions, and only very few destinations are entirely inaccessible within the set 30-minute threshold. It is important to note that since most destinations are close to the neighbourhood, they can comfortably be reached within a half-hour walk and do not require travel by public transport. The commuter rail services extend the area reachable along the railway corridor. This is clearest along the south-western railway corridor, where stops are densely located, and services are more frequent than on the eastern corridor.

Figure 8 shows dynamic public transport accessibility during a typical workday from the Priisle neighbourhood within a 30-minute journey. In comparison to the Kalamaja neighbourhood, there is a substantially more significant proportion of destinations not accessible by public transport within the set threshold. 64.2% of observed destinations are accessible at least once within half an hour, and only 33.2% are consistently accessible throughout the day. It is also clearly visible that the public transport network provides much better connectivity towards the city centre than other surrounding areas. However, 35.8% of destinations are not accessible within

Table 3 Access to observed destinations within a 30-minute range by public transport or walking, per case study area

Access level	Number of trips from Kalamaja	Number of trips from Priisle
5 (> 80%)	1,215 (74.6%)	1,125 (33.2%)
4 (> 60%)	19 (1.2%)	104 (3.1%)
3 (> 40%)	49 (3.0%)	451 (13.3%)
2 (> 20%)	27 (1.7%)	61 (1.8%)
1 (> 0.1%)	98 (6.0%)	434 (12.8%)
0 (< 0.1%)	220 (13.5%)	1,214 (35.8%)

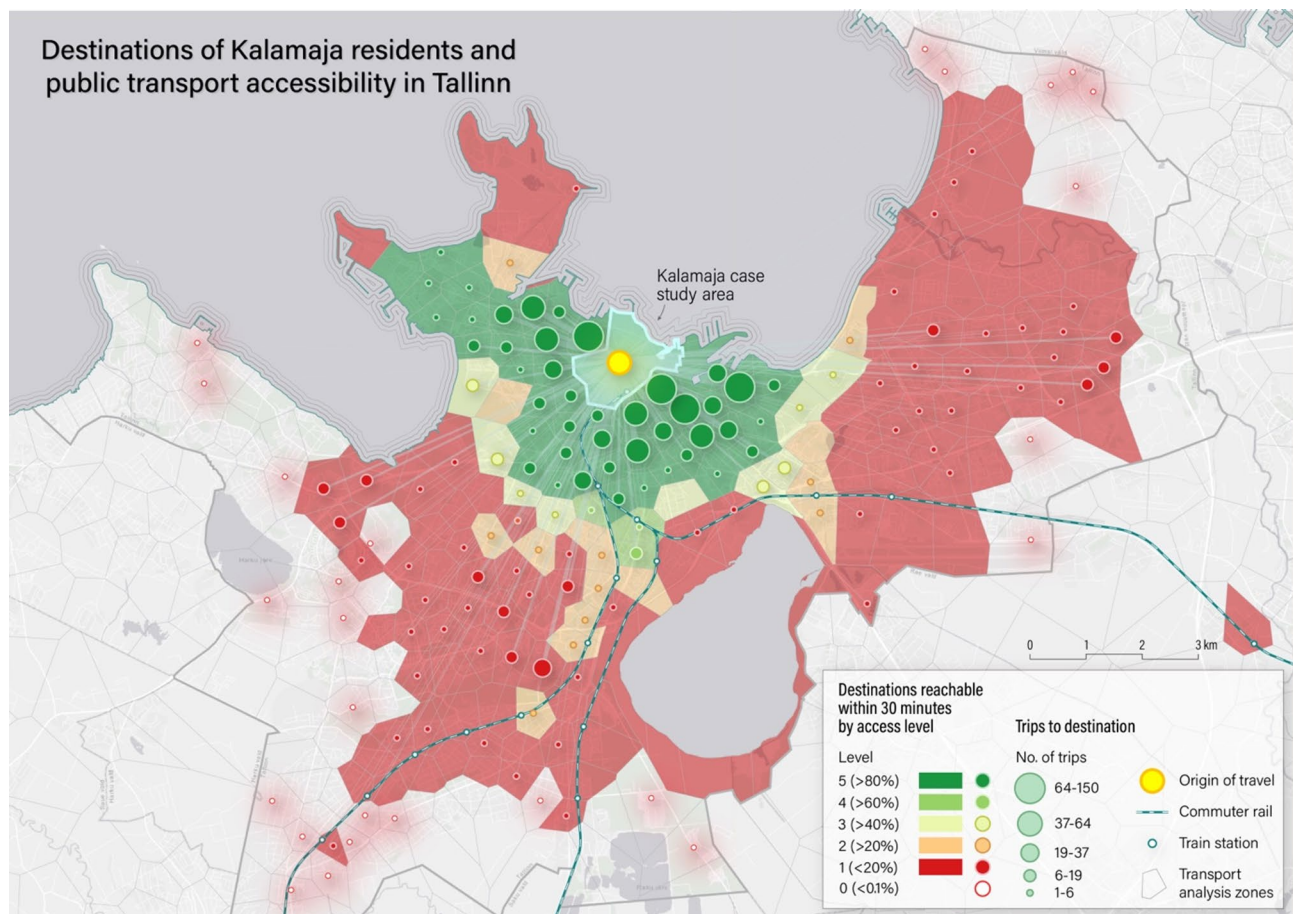


Fig. 7 Dynamic public transport accessibility to observed destinations from Kalamaja

half an hour, most of which are located on the other side of the city centre. There are sharp drops in the level of access around the case study area of Priisle. This can be explained by the lack of public transport services in these directions and natural barriers such as cliffs and rivers around the neighbourhood, limiting access on foot.

Comparing the two neighbourhoods, it is evident that residents of Kalamaja have much higher and much more consistent levels of access to desired destinations throughout the day than residents of Priisle. On average, 77.3% of destinations are accessible to Kalamaja residents throughout the day, with a low standard deviation of 2.7%. In contrast, Priisle residents have access to only 43.2% of destinations on average, with a higher standard deviation of 7.2% (Fig. 9). The distribution of distances to these destinations also varies notably between the neighbourhoods. The median distance to destinations is 1.9 km for Kalamaja residents and 5.3 km for Priisle residents (Fig. 10).

Visualising the areas accessible within a 30-minute journey at various times of day highlights the differences in access by public transport and walking (Fig. 11). The areas accessible are noticeably larger during peak times,

such as 8:15 and 18:00, in comparison to off-peak times like 11:00 and 21:00.

9 Discussion

Enhancing public transport access to desired destinations observed from long-term travel patterns is critical in improving public transport service. This helps increase social equity and the use of public transport and other sustainable travel modes, consequently reducing car dependence in cities. In this study, we analysed three months of GPS data from residents in two neighbourhoods in Tallinn to identify predominant travel destinations. We then assessed the spatial accessibility provided by the public transport system in each neighbourhood and analysed how access to destinations fluctuated throughout the day. Our findings highlighted areas where transport services could be improved to better align with observed travel behaviour.

10 Dynamic public transport accessibility

The results of our analysis emphasise the importance of incorporating the temporal dimension when assessing public transport accessibility, as it reveals how the public

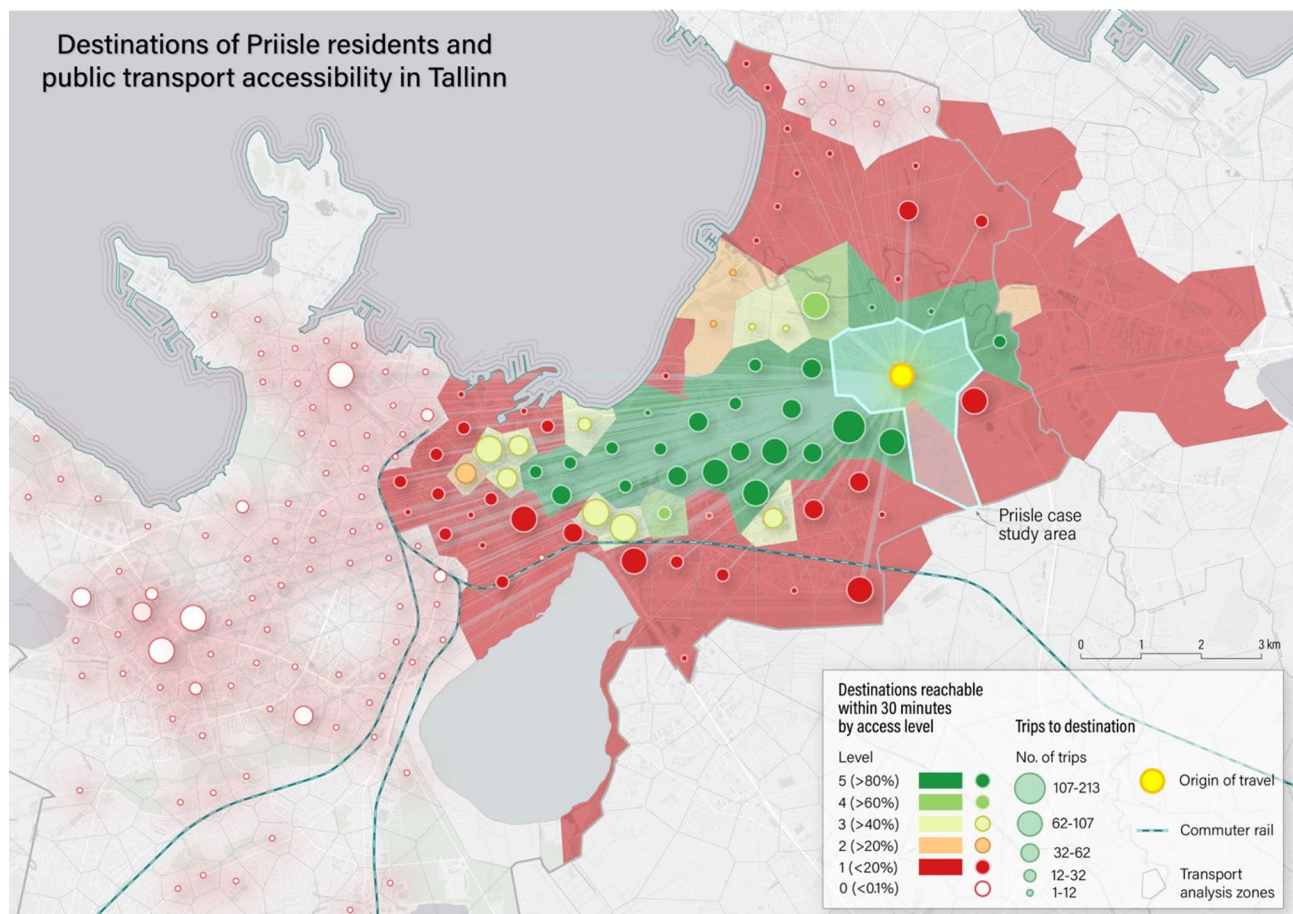


Fig. 8 Dynamic public transport accessibility to observed destinations from Priisle

transport network structure and scheduling affect the range of destinations typically reachable within a fixed travel time budget. Clear differences in destination accessibility are noticeable between the neighbourhoods of Kalamaja and Priisle across different times of day.

Residents of the inner-city neighbourhood of Kalamaja could reach an average of 77.3% of desired destinations within 30 min by public transport and walking throughout the day. Accessibility remained relatively stable across times of day, as most destinations were nearby, with a median distance of just 1.9 km. Almost two-thirds (65.5%) of trips from Kalamaja covered short distances (<2.25 km), making them reachable within half an hour at a walking speed of 4.5 km/h.

In contrast, the inner suburb of Priisle had significantly lower levels of access to destinations by public transport and walking. Destinations were much further away, with a median distance of 5.3 km. Consequently, only an average of 43.2% of destinations could be reached within a 30-minute journey throughout the day. The standard deviation of 7.2% further shows how public transport service patterns at different times of day have a more profound impact on accessibility for Priisle residents than

those of Kalamaja. Located near Tallinn's administrative border, Priisle is primarily served by public transport routes oriented westward toward the city centre, which limits the accessibility to destinations in other surrounding directions.

11 Methodological considerations and limitations

Using mobile phone-based GPS data from study participants to detect travel patterns presents several challenges, such as population bias, privacy concerns, participant recruitment, and complexities of data pre-processing [20]. One common limitation of GPS-based studies is the typically limited sample size, as it is challenging to recruit a large, representative sample for long-term mobile phone tracking [45]. We used a carefully constructed sample based on socio-economic quotas to ensure diverse representation [37]. However, the final sample did not meet all quotas set due to sample recruitment challenges for mobile phone tracking over a longer time. Comprehensive public transport planning for increased social equity ideally requires data that captures all age groups, which our sample could not fully represent. Furthermore, due to the small sample size, we

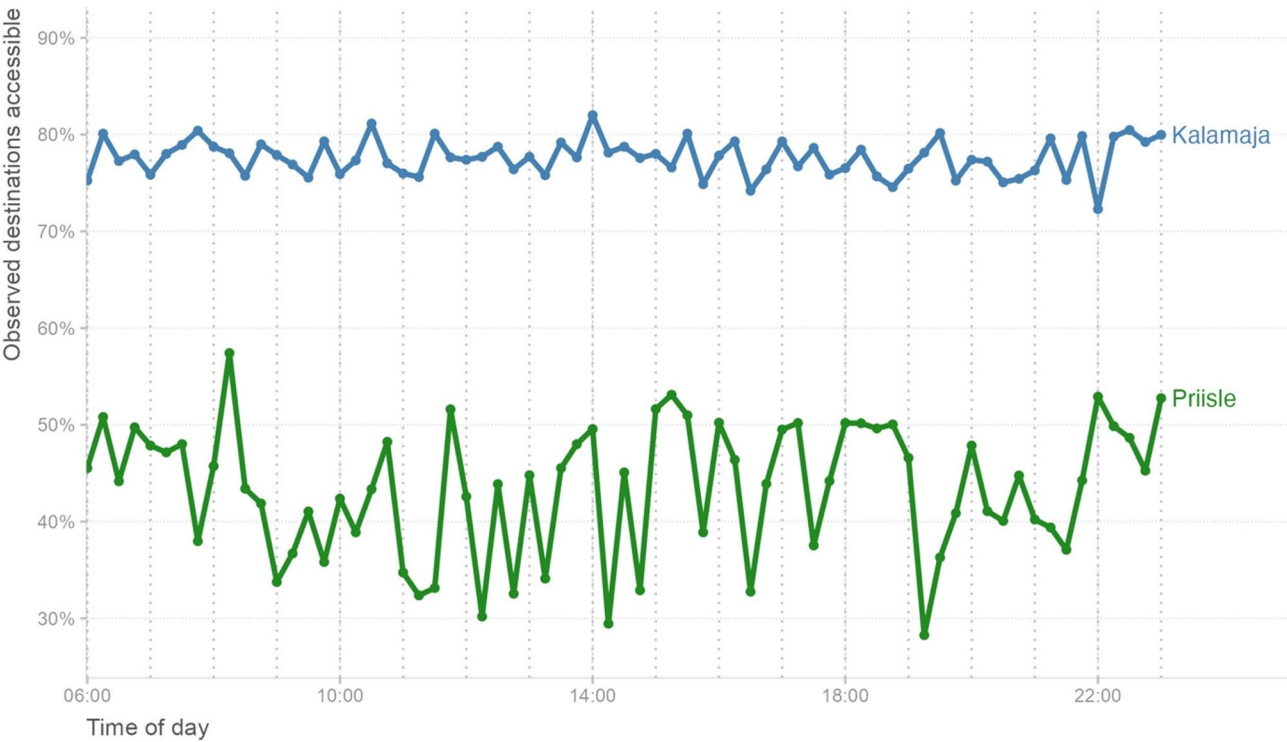


Fig. 9 Percentage of desired destinations accessible by public transport and walking from the neighbourhoods of Kalamaja and Priisle at different times of day

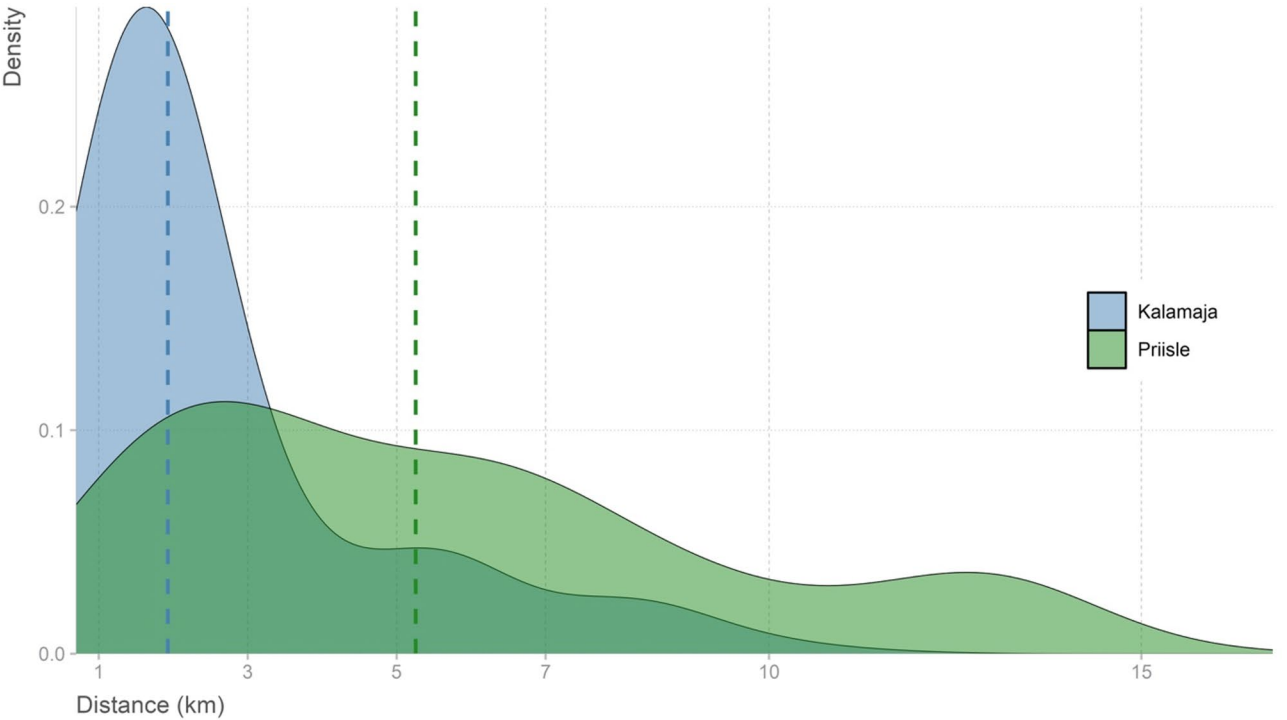


Fig. 10 Distribution of the Euclidean distances of destinations from Kalamaja and Priisle. The vertical dotted lines show the median value of either neighbourhood

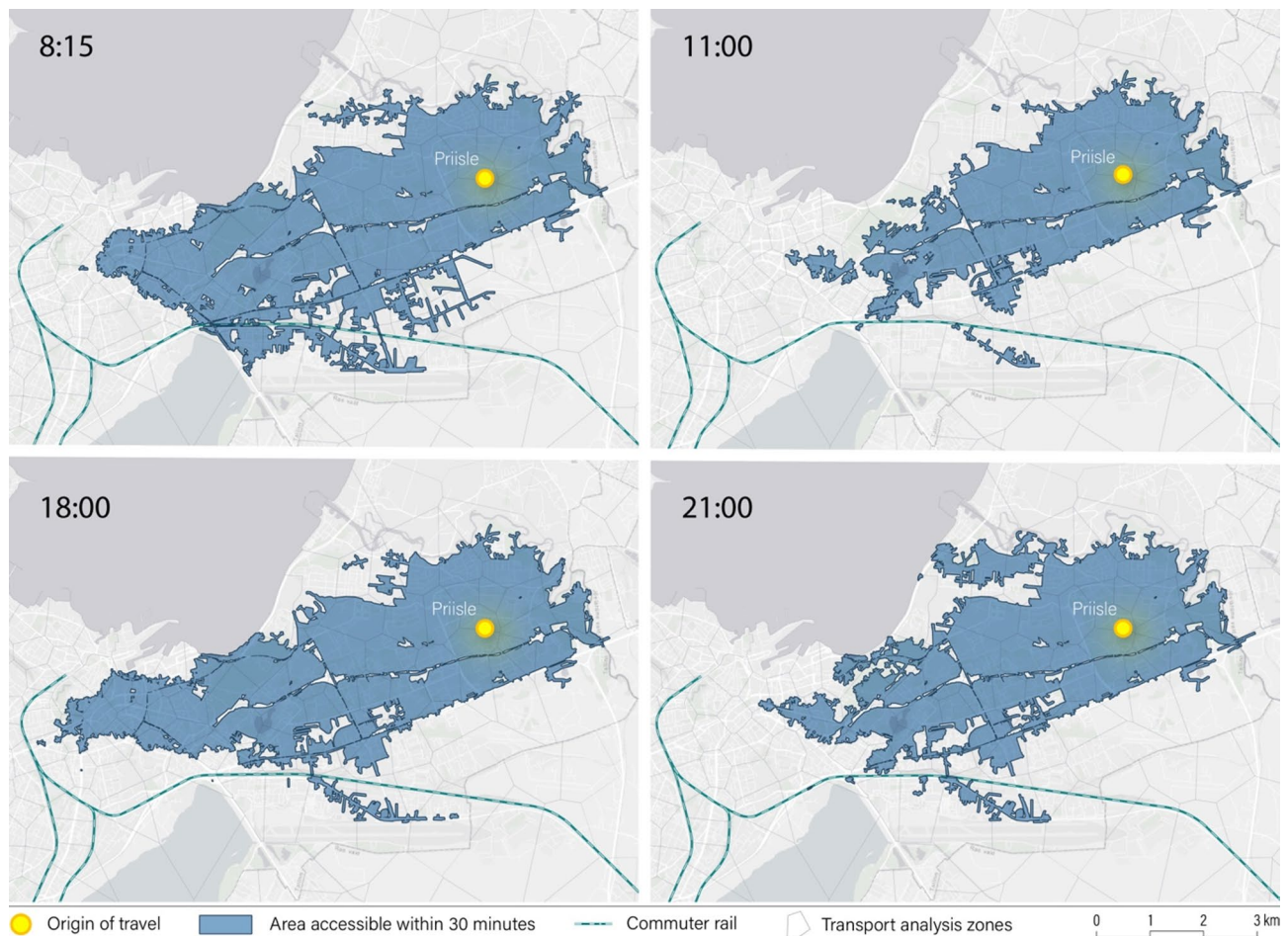


Fig. 11 Areas accessible within a 30-minute journey from Priisle neighbourhood at different times of day by public transport and walking

could not analyse variations in travel destinations across different times of day or incorporate these insights into our dynamic analysis of public transport accessibility. Expanding the sample to include a broader demographic range and participants from all neighbourhoods in the city would enable conducting a similar analysis on a larger scale. This would allow for a more detailed study of the mismatch between travel behaviour and the accessibility provided by public transport. Nonetheless, the data collected in the study provided valuable insights into long-term travel patterns among the working-age population.

Our approach of dividing continuous GPS data into two-minute segments for trip chain detection has notable advantages. Such segmentation reduces data processing time and volume, as GPS points are classified as either movement or stop episodes based on aggregated segments rather than individual points. This is beneficial when working with extensive, long-term GPS datasets demanding significant processing, user engagement, and computational power [16].

We calculated the areas accessible by public transport and walking using a fixed walking speed of 4.5 km/h. However, studies have shown that walking speeds can vary significantly depending on the person's physical capabilities and seasonal factors [47], meaning the areas accessible within a certain travel time may differ between individuals. Additionally, GTFS data relies on planned public transport schedules and does not account for delays or disruptions, leading to overestimating actual accessibility by 4–15% [31, 46]. Supplementing GTFS data with long-term average GPS data from sensors on public transport vehicles could improve the accuracy of in-vehicle journey times. Furthermore, our model does not limit the number of transfers within a single trip, which may result in unrealistic or impractical travel scenarios.

Although GPS data provides rich detail on movement patterns, our analysis focused on identifying trip destinations rather than the precise route, trip purpose, or travel mode. While integrating supplementary data sources, such as travel diaries or surveys, could enhance the accuracy of trip chain detection and help better understand the observed travel behaviour [32, 48], our analysis

showed that detecting trip destinations solely from continuous GPS data was sufficient for dynamic public transport accessibility assessment.

COVID-19 restrictions were minimal during our study period, so their impact on mobility could be considered negligible. Despite these limitations, our analysis effectively demonstrates the value of mobile phone-based GPS data in capturing long-term travel patterns for dynamically assessing public transport accessibility to desired destinations.

12 Policy implications and future research

Our findings have several noteworthy takeaways from a public policy perspective. The structure of the public transport network and the scheduling of individual lines have a large impact on the accessibility of destinations. Reducing overall journey times is crucial to enhancing the attractiveness of public transport relative to car use, as shorter travel times increase the number of accessible destinations within half an hour. For example, journeys requiring transfers introduce additional delays, as passengers experience both transfer and waiting times. This makes it challenging to reach destinations within half an hour. Frequent and direct services segregated from other travel modes to key destinations can increase the mode share of public transport and help achieve the transition to sustainable mobility. Identifying prevalent destinations from GPS travel data helps practitioners plan effective public transport services.

A study conducted by [37] on dominant mobility styles in the neighbourhoods of Kalamaja and Priisle revealed that 38% of Priisle residents exhibited a car-dominant style, compared to only 13% in Kalamaja. This is consistent with our findings, as Priisle has much lower levels of accessibility to desired destinations via public transport and walking, making car travel more appealing. Improving public transport options in Priisle, such as through commuter rail, light rail, or bus rapid transit, would extend the range of destinations reachable within 30 min. Kalamaja residents already benefit from enhanced accessibility to areas served by commuter rail services, emphasising the value of these options in increasing access to destinations.

13 Conclusions

Transitioning to sustainable urban mobility and reducing car dependency necessitates an equitable public transport system that ensures high access levels. Our research highlights the value of using long-term GPS data in evaluating public transport accessibility to observed travel destinations. We introduced a novel framework for trip chain detection from continuous GPS data to identify long-term travel patterns and desired destinations. We then dynamically assessed public transport accessibility

to the observed destinations, accounting for temporal variability throughout the day in Tallinn, Estonia.

Our findings revealed significant differences in accessibility between the two case study neighbourhoods of Kalamaja and Priisle in Tallinn. Kalamaja residents benefited from higher and more consistent levels of public transport accessibility to desired destinations throughout the day, primarily due to the proximity of destinations and the availability of public transport services. In contrast, Priisle residents faced significantly lower accessibility levels, with over a third of all desired destinations being always unreachable within a 30-minute journey by public transport and walking. The noticeable variations in accessibility throughout the day for Priisle residents further demonstrated the importance of dynamic accessibility analysis.

The results of our research emphasise the need for transport planners to consider both spatial and temporal dimensions when measuring accessibility. Providing higher levels of accessibility through more frequent and direct public transport services to key destinations, particularly in underserved areas like Priisle, can mitigate social exclusion and promote a shift towards sustainable travel modes. Future research should focus on incorporating larger samples to include a broader demographic range and integrating supplementary data sources to refine trip chain detection and accessibility assessments further.

This work is significant in the field of public transport accessibility assessment as it is among the few that base the analysis on long-term GPS data for identifying the mobility patterns of urban dwellers and account for spatial variations in access depending on the time of day.

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Author contributions

Martin Haamer conceptualized the study, designed the methodology, conducted the data analysis, wrote the draft of the manuscript and led the research project. Anto Aasa contributed to the study design, methodology design, data interpretation, and interpretation of results. Age Poom contributed to the study design, reviewed and edited the manuscript, and provided critical feedback. All authors read and approved the final manuscript. Each of the authors confirms that this manuscript has not been previously published and is not currently under consideration by any other journal.

Data availability

The GPS-based data used in our study contain sensitive personal information and is not publicly available.

Declarations

Competing interests

The authors declare that they have no competing interests.

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References

- Ahas, R., Aasa, A., Silm, S., & Tiru, M. (2010). Daily rhythms of suburban commuters' movements in the Tallinn metropolitan area: Case study with mobile positioning data. *Transportation Research Part C: Emerging Technologies*, 18(1), 45–54. <https://doi.org/10.1016/j.trc.2009.04.011>
- Albacete, X., Olaru, D., Paül, V., & Biermann, S. (2017). Measuring the accessibility of public transport: A critical comparison between methods in Helsinki. *Applied Spatial Analysis and Policy*, 10(2), 161–188. <https://doi.org/10.1007/s12061-015-9177-8>
- Allen, J., & Farber, S. (2020). Planning transport for social inclusion: An accessibility-activity participation approach. *Transportation Research Part D: Transport and Environment*, 78, 102212. <https://doi.org/10.1016/j.trd.2019.102212>
- Barbosa, H., Barthelemy, M., Ghoshal, G., James, C. R., Lenormand, M., Louail, T., Menezes, R., Ramasco, J. J., Simini, F., & Tomasini, M. (2018). Human mobility: Models and applications. *Physics Reports*, 734, 1–74. <https://doi.org/10.1016/j.physrep.2018.01.001>
- Bejleri, I., Noh, S., Gu, Z., Steiner, R. L., & Winter, S. M. (2018). Analytical method to determine transportation service gaps for transportation disadvantaged populations. *Transportation Research Record: Journal of the Transportation Research Board*, 2672(8), 649–661. <https://doi.org/10.1177/0361198118794290>
- Chen, C., Ma, J., Susilo, Y., Liu, Y., & Wang, M. (2016). The promises of big data and small data for travel behavior (aka human mobility) analysis. *Transportation Research Part C: Emerging Technologies*, 68, 285–299. <https://doi.org/10.1016/j.trc.2016.04.005>
- City of Tallinn (2023). Tallinna jätkusuutliku linnaliikuvuse kava. <https://www.tallinn.ee/et/media/525704>
- City of Tallinn (2020). Tallinna arengu strateegia „Tallinn 2035“. <https://www.riigit.eataja.ee/akt/429122020009>
- Dalumpines, R., & Scott, D. M. (2017). Making mode detection transferable: Extracting activity and travel episodes from GPS data using the multinomial logit model and Python. *Transportation Planning and Technology*, 40(5), 523–539. <https://doi.org/10.1080/03081060.2017.1314502>
- EEA (2022). *Transport and environment report 2021* (Publication No. 02/2022). European Environment Agency. <https://www.eea.europa.eu/publications/transport-and-environment-report-2021>
- ESRI (2024). Public-transit-tools/transit-network-analysis-tools/README.md at master · Esri/public-transit-tools. GitHub. <https://github.com/Esri/public-transit-tools/blob/master/transit-network-analysis-tools/README.md>
- Fan, J., Meng, X., Tian, J., Xing, C., Wang, C., & Wood, J. (2023). A review of transportation carbon emissions research using bibliometric analyses. *Journal of Traffic and Transportation Engineering (English Edition)*, 10(5), 878–899. <https://doi.org/10.1016/j.jtte.2023.09.002>
- Fayyaz, S. K., Liu, X. C., & Porter, R. J. (2017). Dynamic transit accessibility and transit gap causality analysis. *Journal of Transport Geography*, 59, 27–39. <https://doi.org/10.1016/j.jtrangeo.2017.01.006>
- Fink, C., Willberg, E., Klein, R., Heikinheimo, V., & Toivonen, T. (2024). A travel time matrix data set for the Helsinki region 2023 that is sensitive to time, mode and interpersonal differences, and uses open data and novel open-source software. *Scientific Data*, 11(1), 858. <https://doi.org/10.1038/s41597-024-03689-z>
- Fransen, K., Neutens, T., Farber, S., De Maeyer, P., Deruyter, G., & Witlox, F. (2015). Identifying public transport gaps using time-dependent accessibility levels. *Journal of Transport Geography*, 48, 176–187. <https://doi.org/10.1016/j.jtrangeo.2015.09.008>
- Gadziński, J. (2018). Perspectives of the use of smartphones in travel behaviour studies: Findings from a literature review and a pilot study. *Transportation Research Part C: Emerging Technologies*, 88, 74–86. <https://doi.org/10.1016/j.trc.2018.01.011>
- Geurs, K. T., & Reggiani, A. (2014). Recent advances and applications in accessibility modelling. *Computers Environment and Urban Systems*, 49. <https://doi.org/10.1016/j.compenvurbsys.2014.09.003>
- Goliszek, S., Polom, M., & Duma, P. (2020). Potential and cumulative accessibility of workplaces by public transport in Szczecin. *Bulletin of Geography Socio-Economic Series*, 50, Article 50. <https://doi.org/10.2478/bog-2020-0037>
- Hansen, W. G. (1959). How accessibility shapes land use. *Journal of the American Institute of Planners*, 25(2), 73–76. <https://doi.org/10.1080/01944365908978307>
- Harrison, G., Grant-Muller, S. M., & Hodgson, F. C. (2020). New and emerging data forms in transportation planning and policy: Opportunities and challenges for track and trace data. *Transportation Research Part C: Emerging Technologies*, 117, 102672. <https://doi.org/10.1016/j.trc.2020.102672>
- Hluško, R., Stanek, R., Ďurček, P., & Kusendová, D. (2024). Urban public transport system accessibility for different groups of residents: Case of Bratislava City. *Case Studies on Transport Policy*, 16, 101200. <https://doi.org/10.1016/j.cstp.2024.101200>
- ITF (2017). The economic benefits of improved accessibility to transport systems. *OECD*. <https://doi.org/10.1787/9c73ac17-en>
- Järvi, O., Tenkanen, H., Salonen, M., Ahas, R., & Toivonen, T. (2018). Dynamic cities: Location-based accessibility modelling as a function of time. *Applied Geography*, 95, 101–110. <https://doi.org/10.1016/j.apgeog.2018.04.009>
- Kaeoruean, K., Phithakitnukoon, S., Demissie, M. G., Kattan, L., & Ratti, C. (2020). Analysis of demand–supply gaps in public transit systems based on census and GTFS data: A case study of Calgary, Canada. *Public Transport*, 12(3), 483–516. <https://doi.org/10.1007/s12469-020-00252-y>
- Kalvo, R., Haapamäki, T., Krause, C., Mäkinen, S., Ramseier, R., & Rannala, M. (2021). Tallinna regiooni liikuvuse mudeldamine. Grupp trükiagentuur OÜ. http://transpordiamet.ee/sites/default/files/documents/2022-02/Tallinna%20regiooni%20liikuvuse%20mudeldamine_20210705E.pdf
- Kapatsila, B., Palacios, M. S., Grisé, E., & El-Genaidy, A. (2023). Resolving the accessibility dilemma: Comparing cumulative and gravity-based measures of accessibility in eight Canadian cities. *Journal of Transport Geography*, 107, 103530. <https://doi.org/10.1016/j.jtrangeo.2023.103530>
- Küpper, A. (2005). *Location-Based Services: Fundamentals and Operation* (1st ed.). Wiley. <https://doi.org/10.1002/0470092335>
- Lei, T. L., & Church, R. L. (2010). Mapping transit-based access: Integrating GIS, routes and schedules. *International Journal of Geographical Information Science*, 24(2), 283–304. <https://doi.org/10.1080/13658810902835404>
- Mavoa, S., Witten, K., McCreanor, T., & O'Sullivan, D. (2012). GIS based destination accessibility via public transit and walking in Auckland, New Zealand. *Journal of Transport Geography*, 20(1), 15–22. <https://doi.org/10.1016/j.jtrangeo.2011.10.001>
- Neutens, T., Schwanen, T., Witlox, F., & De Maeyer, P. (2010). Equity of urban service delivery: A comparison of different accessibility measures. *Environment and Planning A: Economy and Space*, 42(7), 1613–1635. <https://doi.org/10.1068/a4230>
- Nichols, A., Ryan, J., & Palmqvist, C. W. (2024). The importance of recurring public transport delays for accessibility and mode choice. *Journal of Transport Geography*, 115, 103796. <https://doi.org/10.1016/j.jtrangeo.2024.103796>
- Nitsche, P., Widhalm, P., Breuss, S., Brändle, N., & Maurer, P. (2014). Supporting large-scale travel surveys with smartphones – A practical approach. *Transportation Research Part C: Emerging Technologies*, 43, 212–221. <https://doi.org/10.1016/j.trc.2013.11.005>
- Pereira, R. H. M., Schwanen, T., & Banister, D. (2017). Distributive justice and equity in transportation. *Transport Reviews*, 37(2), 170–191. <https://doi.org/10.1080/01441647.2016.1257660>
- Poom, A. (2019). Tegevusruumiuringud GPSi andmete abil: Mobiilsuuringute labori andmeinfrastruktuur. In T. Pae, & Ü. Mander (Eds.), *Publications Instituuti Geographici Universitatis Tartuens: Uurimusi eestikeelse geograafia 100. aastapäeval* (Vol. 113, pp. 204–214). Tartu Ülikooli Kirjastus.
- Salonen, M., & Toivonen, T. (2013). Modelling travel time in urban networks: Comparable measures for private car and public transport. *Journal of Transport Geography*, 31, 143–153. <https://doi.org/10.1016/j.jtrangeo.2013.06.011>
- Schuessler, N., & Axhausen, K. W. (2009). Processing Raw data from global positioning systems without additional information. *Transportation Research Record*, 2105(1), 28–36. <https://doi.org/10.3141/2105-04>
- Silm, S., Tominga, A., Saidla, K., Poom, A., & Tammaru, T. (2024). Socio-economic and residential differences in urban modality styles based on a long-term smartphone experiment. *Journal of Transport Geography*, 115, 103810. <https://doi.org/10.1016/j.jtrangeo.2024.103810>
- Solmaz, G., & Turgut, D. (2019). A Survey of Human Mobility Models. *IEEE Access*, 7, 125711–125731. <https://doi.org/10.1109/ACCESS.2019.2939203>
- Statistics Estonia (2024). *RV06U: DEMOGRAPHIC LABOUR PRESSURE INDEX, 1 JANUARY by Year, Region/Administrative unit and Indicator*. Statistical database [Dataset]. https://andmed.stat.ee/en/stat/rhvastik__rhvastikunaitajad-ja-koosseis__demograafilised-pehinaitaajad/RV06U/table/tableViewLayout2

40. Stępnia, M., & Goliszek, S. (2017). *Spatio-Temporal Variation of Accessibility by Public Transport—The Equity Perspective* (pp. 241–261). https://doi.org/10.1007/978-3-319-45123-7_18
41. Stępnia, M., Pritchard, J. P., Geurs, K. T., & Goliszek, S. (2019). The impact of Temporal resolution on public transport accessibility measurement: Review and case study in Poland. *Journal of Transport Geography*, 75, 8–24. <https://doi.org/10.1016/j.jtrangeo.2019.01.007>
42. Tenkanen, H., & Toivonen, T. (2020). Longitudinal Spatial dataset on travel times and distances by different travel modes in Helsinki region. *Scientific Data*, 7(1), 77. <https://doi.org/10.1038/s41597-020-0413-y>
43. Toch, E., Lerner, B., Ben-Zion, E., & Ben-Gal, I. (2019). Analyzing large-scale human mobility data: A survey of machine learning methods and applications. *Knowledge and Information Systems*, 58(3), 501–523. <https://doi.org/10.1007/s10115-018-1186-x>
44. Transport Administration (2024). *Liikuvuse statistika | Transpordiamet*. <https://transpordiamet.ee/liikuvuse-statistika>
45. Wang, Z., He, S. Y., & Leung, Y. (2018). Applying mobile phone data to travel behaviour research: A literature review. *Travel Behaviour and Society*, 11, 141–155. <https://doi.org/10.1016/j.tbs.2017.02.005>
46. Wessel, N., & Farber, S. (2019). On the accuracy of schedule-based GTFS for measuring accessibility. *Journal of Transport and Land Use*, 12(1). <https://doi.org/10.5198/jtlu.2019.1502>
47. Willberg, E., Fink, C., & Toivonen, T. (2023). The 15-minute City for all? – Measuring individual and Temporal variations in walking accessibility. *Journal of Transport Geography*, 106, 103521. <https://doi.org/10.1016/j.jtrangeo.2022.103521>
48. Xiao, G., Juan, Z., & Zhang, C. (2016). Detecting trip purposes from smartphone-based travel surveys with artificial neural networks and particle swarm optimization. *Transportation Research Part C: Emerging Technologies*, 71, 447–463. <https://doi.org/10.1016/j.trc.2016.08.008>
49. Zochowska, R., Kłos, M. J., Soczówka, P., & Pilch, M. (2022). Assessment of accessibility of public transport by using Temporal and Spatial analysis. *Sustainability*, 14(23). <https://doi.org/10.3390/su142316127>. Article 23.

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